

A 2-kV Charge-Based ZVS Three-Level Inverter

Sjef J. Settels , *Student Member, IEEE*, Jorge L. Duarte , *Member, IEEE*,
Jeroen van Duivenbode, *Member, IEEE*, and Elena A. Lomonova, *Senior Member, IEEE*

Abstract—Industrial applications, e.g., semiconductor manufacturing equipment, require power converters providing high power with high precision and bandwidth. This article presents a three-level flying capacitor resonant pole inverter configuration that combines high output power and high switching frequency with reduced switch voltage stress. A multilevel modulation strategy is applied to minimize conduction losses, which, in addition, guarantees zero-voltage switching (ZVS) for the entire operating range to reduce switching losses. The proposed multilevel converter configuration is compared with an existing two-level configuration by simulation. Increasing the number of voltage levels results in lower total losses and increased linearity of the generated output current. Experimental results acquired with a hardware prototype validate the fast switching of high voltage, proper functioning of the multilevel modulation strategy, and achieving of ZVS. Improvements of the theoretical analysis are presented to compensate for delays in the system and deviating parameter values. Results obtained with a compensated system indicate a relatively high accuracy and linearity of the generated output current of the inverter.

Index Terms—High voltage power converters, modulation strategy, multilevel converters, resonant converter, soft switching.

I. INTRODUCTION

EQUIPMENT used in semiconductor manufacturing applies high-precision fast-moving stages for accurate positioning of wafers to fulfill tasks such as exposure, inspection, or dicing [1], [2] (see Fig. 1 [3]). The required positioning accuracy is in the nanometer range, which results in stringent requirements on the output current accuracy and bandwidth of the power converters driving the various types of actuators involved [4], [5]. At the same time, semiconductor manufacturers strive to minimize machine operating cost by maximizing machine throughput. As a result, the demand for mechanical power, and therefore, electrical power, is evermore increasing. Since the actuators of the moving stages operate in a dynamic environment, it is desirable to keep the weight and rigidity of the actuator cables low as the resulting disturbance forces are difficult to model and compensate. Therefore, increasing processed power by increasing operating voltage is favorable, compared to increasing operating current, which the converter should be capable of providing.

Manuscript received December 22, 2018; revised March 26, 2019 and May 20, 2019; accepted August 7, 2019. Date of publication August 13, 2019; date of current version January 10, 2020. Recommended for publication by Associate Editor T. Mishima. (Corresponding author: Sjef J. Settels.)

The authors are with the Department of Electrical Engineering, Eindhoven University of Technology, 5612 AZ Eindhoven, The Netherlands (e-mail: s.settels@tue.nl; j.l.duarte@tue.nl; j.v.duivenbode@tue.nl; e.lomonova@tue.nl).

Color versions of one or more of the figures in this article are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TPEL.2019.2935297

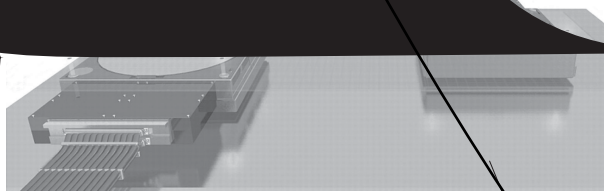
Current power converters used in semiconductor manufacturing equipment operate, for instance using a bus voltage of around 600 V and employ MOSFETs as switching devices. A resonant pole inverter topology using variable hysteresis control is implemented resulting in soft-switching behavior with a switching frequency range of 10–100 kHz [9]–[11]. In order to meet the requirements for future generation semiconductor manufacturing equipment, a bus voltage of 2 kV and switching frequency of ≥ 50 kHz across the entire operating range is proposed. Due to the stringent requirements on output current accuracy for high-precision industrial applications [2], [4], [5], [12], the required switching frequency of the power converter is several orders higher than that applied in common HV applications, such as electrical propulsion and grid-connected converters. This, however, imposes challenges for the power converter as reliable fast-switching devices are not available with blocking voltages ranging up to 2 kV [13].

Previous research conducted in this area has shown that the flying capacitor resonant pole inverter (FC RPI) is a suitable topology to reduce voltage stress on switches through a multilevel configuration [6]–[8], see Fig. 2. This enables the use of fast switches with a voltage rating lower than the bus voltage by dividing the voltage stress over multiple switches. The addition of two switches and a flying capacitor to a standard half-bridge results in a three-level converter. The presented half-bridge is often applied in two or three branches to drive single- or three-phase actuators, and such implementations are covered by this article as well.

Next to a (big) increase in output power, the industry further requires improved power density to not exceed existing weight and volume restrictions. Therefore, an increase in efficiency has to be realized. This can be achieved by actively using the available middle voltage level of the flying capacitor to shape the filter inductor current i_{Lf} , as was previously presented in [8]. With three different gradients for the filter inductor current, a trapezoidal shape is obtained, which has lower rms and peak values. A lower rms value of the current results in lower conduction losses in the switches and filter inductor. Lower peak values result in lower switching losses, lower core losses in the filter inductor L_f , and lower losses in the output filter capacitor C_f .

Previous implementations of the RPI topology applied additional resonant capacitors parallel to the switches and hysteresis current control to achieve zero-voltage switching (ZVS) across the entire operating range [7], [9]–[11], [14]. A more elaborate charge-based modulation strategy for a two-level converter discarding the additional resonant capacitors is proposed in [15]. Piecewise linear approximation of the filter inductor current is

Actuators



Magnet array

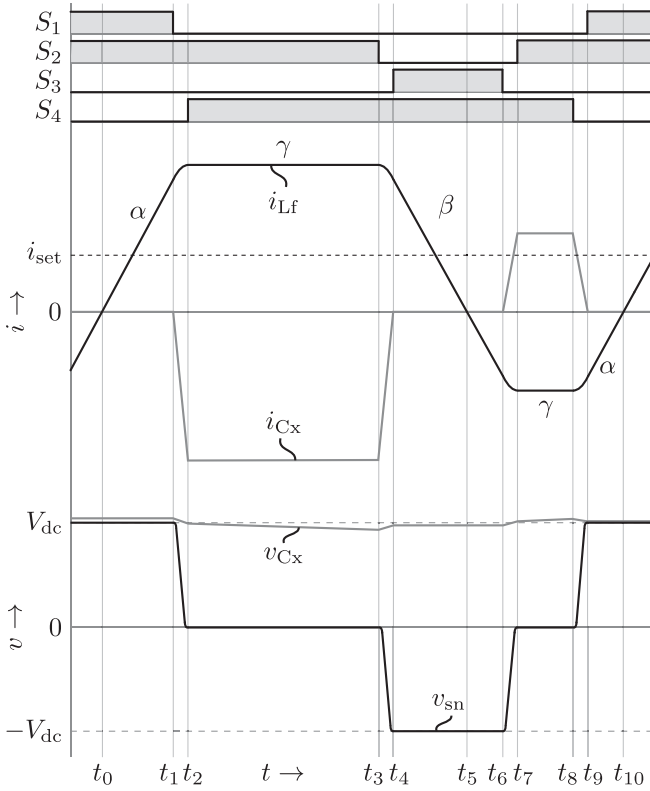


Fig. 3. Schematic representation of the i_{Lf} , i_{Cx} , v_{Cx} , and v_{sn} waveforms for a single switching period. The corresponding drive signals for switches S_{1-4} are shown at the top of the figure. The desired average current for this exemplary schematic representation is $i_{set} > 0$ A, and $v_{out} = 0$ V.

TABLE II
FC RPI SWITCHING STATES

interval	Floating C_x				Charge C_x				Discharge C_x			
	S_1	S_2	S_3	S_4	S_1	S_2	S_3	S_4	S_1	S_2	S_3	S_4
$[t_0, t_1]$	1	1	0	0								
$[t_1, t_2]$					1	0	0	0	0	1	0	0
$[t_2, t_3]$					1	0	1	0	0	1	0	1
$[t_3, t_4]$					0	0	1	0	0	0	0	1
$[t_4, t_5]$	0	0	1	1								
$[t_5, t_6]$	0	0	1	1								
$[t_6, t_7]$					0	0	0	1	0	0	1	0
$[t_7, t_8]$					0	1	0	1	1	0	1	0
$[t_8, t_9]$					0	1	0	0	1	0	0	0
$[t_9, t_{10}]$	1	1	0	0								

filter capacitor, and therefore, increase the converter's efficiency. Fig. 3 shows exemplary waveforms for one switching period of the filter inductor current i_{Lf} , flying capacitor current i_{Cx} and voltage v_{Cx} , and the voltage v_{sn} from the switch node sn to ground. The desired per-period average value of i_{Lf} , defined as i_{set} , is positive and the output voltage v_{out} is equal to 0 V, resulting in a flat middle part of the trapezoidal current shape. The corresponding switching states that result in the exemplary waveforms are shown at the top of Fig. 3 and given in Table II, where the distinction is made between charging or discharging the flying capacitor C_x . The negative part of i_{Lf} is required for

each switching period to achieve ZVS for all turn-ON events of the switches.

The duration of each switching state can be adjusted to obtain the desired average filter current i_{set} , to minimize the rms value of the current through the filter inductor, to regulate the switching frequency, and to guarantee soft-switching at each switching instant. The latter requires the proper calculation of the corresponding time intervals and current values, in order for v_{sn} to completely commutate to the appropriate voltage level as indicated in Fig. 3 [8].

For each switching period, two time instants are available to choose between charging or discharging the flying capacitor in order to balance v_{Cx} around V_{dc} —at t_1 and t_6 . The decision whether to charge or discharge C_x is based on a measurement of the actual voltage v_{Cx} , resulting in closed-loop control of the flying capacitor voltage. Balancing v_{Cx} to V_{dc} results in the switch node voltages for the respective states shown in Table I, depending on the sign of i_{Lf} . The three states for which v_{sn} is constant result in constant slopes of i_{Lf} which are indicated by α , β , and γ in Fig. 3. The corresponding equations for the slopes are

$$\alpha = \frac{V_{dc} - v_{out}}{L_f} \quad (1)$$

$$\beta = \frac{-V_{dc} - v_{out}}{L_f} \quad (2)$$

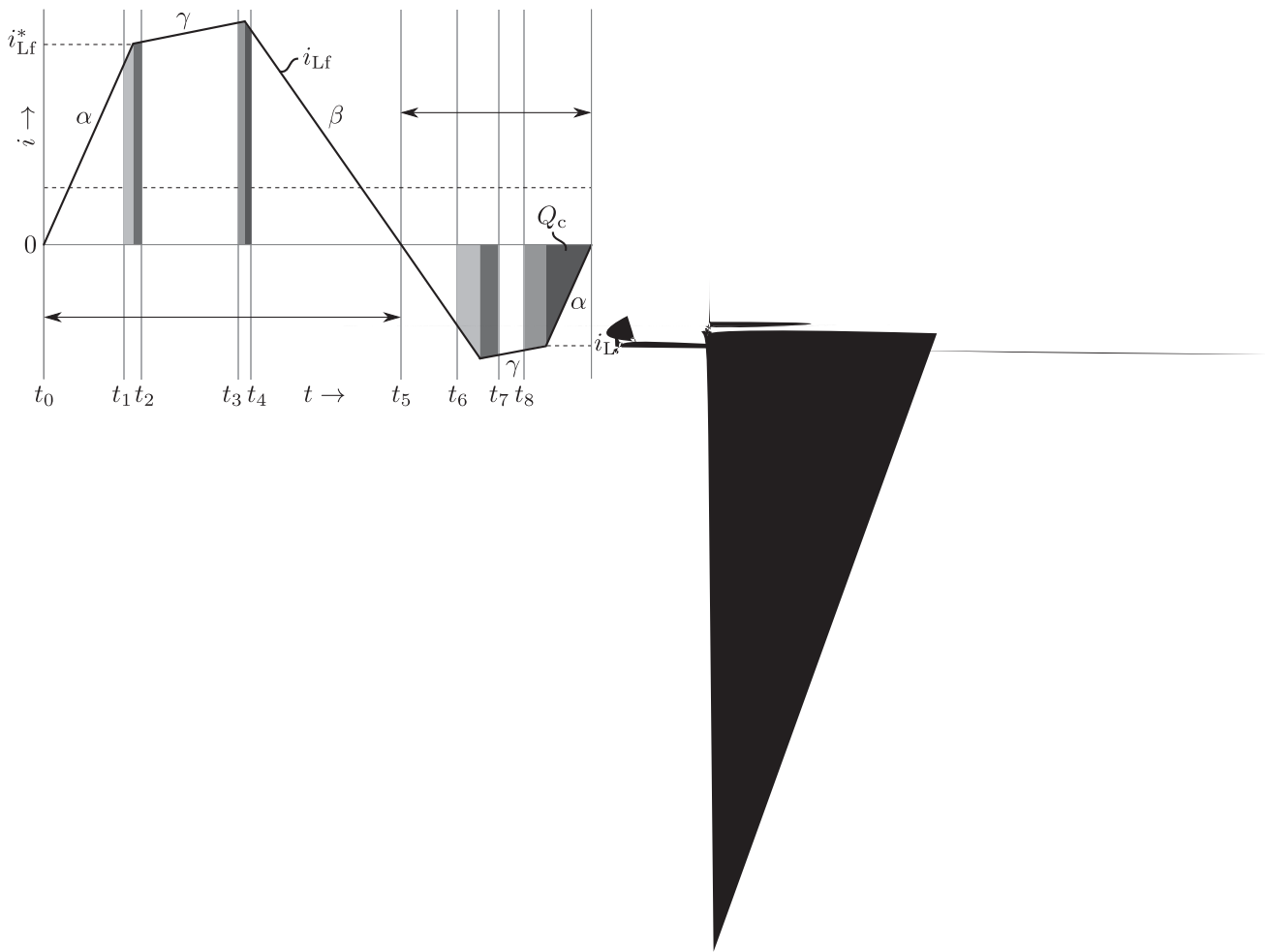
$$\gamma = \frac{v_{sn} - v_{out}}{L_f} \quad (3)$$

Assuming v_{Cx} is balanced properly around V_{dc} with relatively low variations, the approximation can be made that $v_{sn} \approx 0$ V during intervals $[t_2, t_3]$ and $[t_7, t_8]$. However, the flying capacitor is actively delivering a significant amount of energy to the output of the converter. This means that the capacitor has to contain sufficient charge for its ripple voltage $\tilde{v}_{Cx}$ to remain relatively low, and therefore, ensure that the assumption that $v_{sn} \approx 0$ V continues to be valid.

B. Analytical Modeling of Charge-Based ZVS

The nonlinear output capacitance $C_{oss}(v_{ds})$ of a MOSFET can be used as a resonant component for the circulating filter current i_{Lf} and to achieve ZVS in a switching leg [15], [18]. Due to the high nonlinearity of C_{oss} , fast commutation of the switch node voltage is obtained. The charge model of $C_{oss}(v_{ds})$ enables the use of piecewise linear approximation of the trapezoidal shape of i_{Lf} , which is shown in Fig. 4 for $i_{set} > 0$ A and $v_{out} < 0$ V. The charge-based analysis presented in [15] was applied to a three-level FC RPI in [8]. In this article, the basic analysis is extended and an optimization to limit the switching frequency is presented.

To reduce the rms value of i_{Lf} , thereby increasing efficiency, the surface of the negative part of i_{Lf} (indicated with t_N), needed to achieve ZVS, is to be minimized. As a first step, a critical area underneath i_{Lf} is defined which has to contain at least the commutation charge Q_c of the output capacitance $C_{oss}(v_{ds})$ of a single MOSFET, assuming $v_{ds} = V_{dc}$. The critical area is indicated with Q_c in Fig. 4. From this area and the slope of

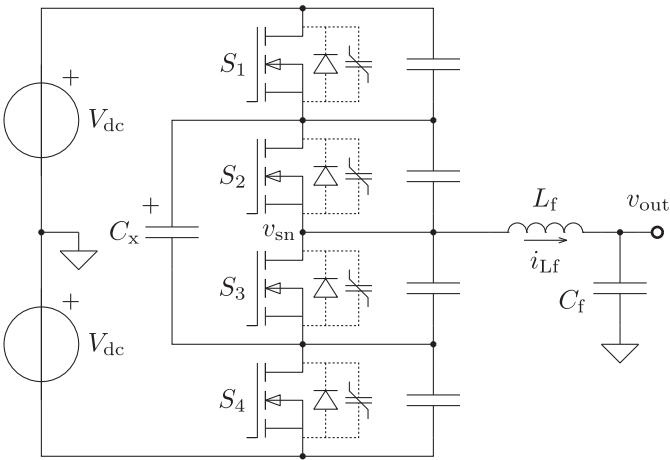


(a)

(b)

(c)

(



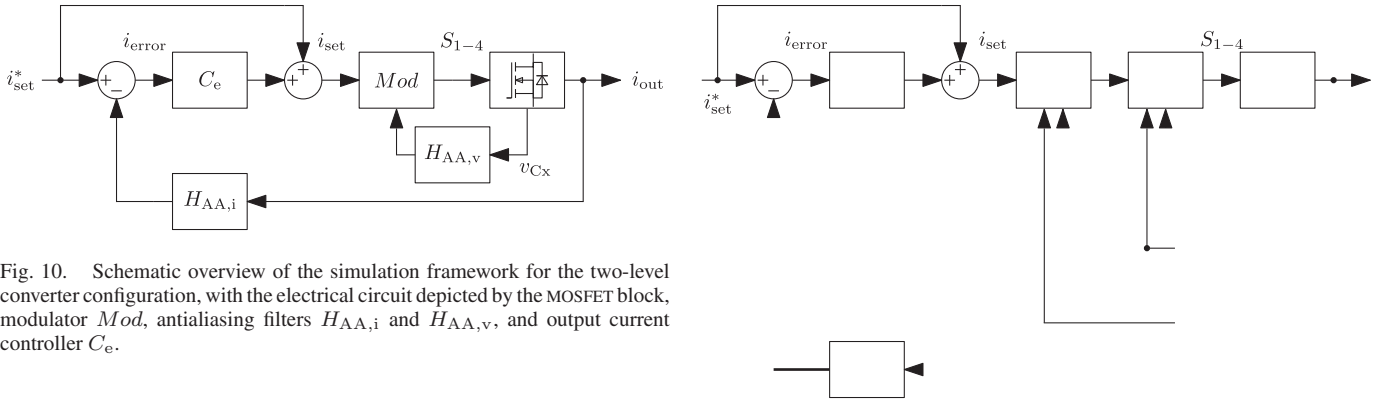


Fig. 10. Schematic overview of the simulation framework for the two-level converter configuration, with the electrical circuit depicted by the MOSFET block, modulator Mod , antialiasing filters $H_{AA,i}$ and $H_{AA,v}$, and output current controller C_e .

TABLE V
SIMULATION SPECIFICATIONS

Parameter	Value	Unit	Parameter	Value	Unit
V_{dc}	1	kV	C_r	820	pF
C_x	60	μF	C_{Qoss}	335	pF
L_f	20	μH	R_o	75	Ω
C_f	20	μF	L_o	5	mH

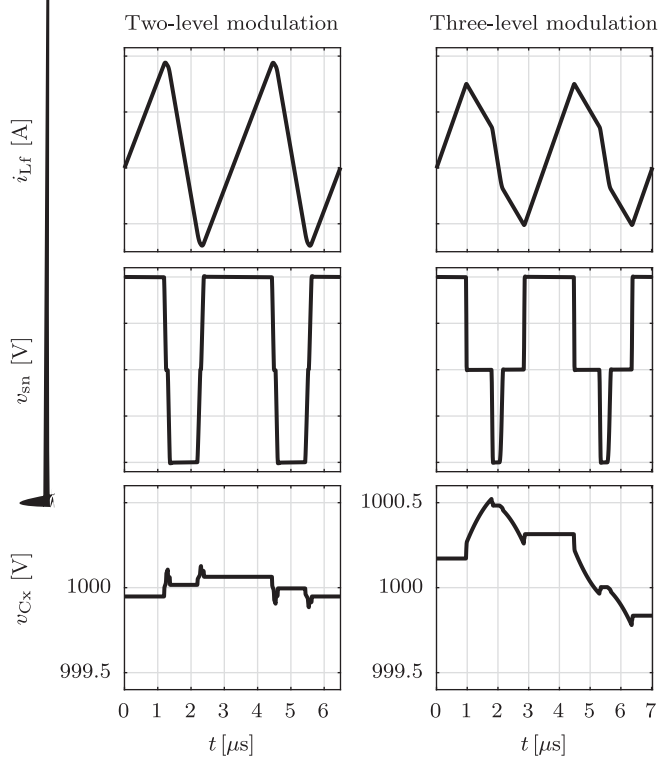
time interval is sufficient to enable active balancing of the flying capacitor voltage for two-level modulation. An accurate determination of the actual flying capacitor voltage v_{Cx} is required and included in the simulation framework.

The resonant pole inverter can be seen as a current controlled converter, for which closed-loop output current control was implemented in the simulation framework [4], [20]. The open-loop bandwidth was tuned to 1 kHz, and antialiasing filters with a cutoff frequency of 100 kHz were added for the measurements of i_{out} and v_{Cx} .

A schematic overview of the implemented simulation framework for the two-level converter configuration is shown in Fig. 10. The switching endstage and electrical circuit are depicted by the MOSFET block which has the gate signals for the four switches S_{1-4} as an input from the modulator Mod . The antialiasing filters are designated $H_{AA,i}$ and $H_{AA,v}$, and the output current controller C_e . The parameters for the simulation framework are given in Table V. The load consists of a series connection of a resistor R_o and inductor L_o , emulating an actuator. The values for L_f , C_r , and C_{Qoss} are chosen in accordance with the values given in [7]. The values for C_x and C_f are equal to the values used for the three-level simulations, which will be elaborated in the next section.

B. Three-Level Simulation Model

For the simulation model of the three-level converter configuration, the electrical circuit diagram shown in Fig. 2 is used. To obtain an equal comparison, the three-level simulation model is based on the same simulation specifications as used for the two-level simulation model, as shown in Table V. The values for L_f and C_x were determined using the reasoning presented in Section II-C and applying worst case parameters determined by simulation. The value of C_f determines the output voltage ripple $\tilde{v}_{out}$ together with L_f , which was chosen $\tilde{v}_{out} \leq 5$ V, from which a minimum value for C_f can be calculated. Furthermore, with the



of the flying capacitor voltages indicate proper balancing of the capacitor voltages to V_{dc} . The difference in voltage ripple resulting from the active use of the additional voltage level for the three-level configuration is shown as well. However, the voltage ripple remains relatively low for the set-point current profile as used in the simulations.

1) *Loss Breakdown Analysis*: A loss breakdown analysis was performed on the obtained results, which incorporates the switching and conduction losses of the switches, core and conduction losses of the filter inductor, and the losses generated in the flying capacitor and filter capacitor.

In order to compare the switching losses $P_{loss,sw}$ resulting from the different modulation strategies, the measurement results presented in [21] and [22] were used to derive an estimate of the losses for the switches used in this article. The results indicate a quadratic relation between the drain-source current at the turn-OFF moment $i_{sw,off}$ and the corresponding switching energy $E_{sw,off}$. A correction factor of 0.5 has been applied to the measured results to compensate for the difference in current handling between the SiC module used in the reference (~ 200 A) and the SiC discrete component used in this article (~ 100 A). This results in the following equation to give an estimation of the turn-OFF switching energy of a single switch:

$$E_{sw,off} = \frac{1}{30} i_{sw,off}^2 [\mu J] \quad (14)$$

where $i_{sw,off}$ is the current through the switch at the turn-OFF moment. The total switching losses for each converter configuration can be calculated by applying (14) for the turn-OFF current of each switch S_{1-4} , assuming ZVS is achieved and negligible turn-ON losses are generated. An estimate of the $R_{ds,on}$ taken from the datasheet of the SiC discrete MOSFET used (Wolfspeed C2M0025120D [19]), and applied together with the currents through the switches, was used to determine the total conduction losses $P_{loss,cond}$ in the switches.

To obtain estimates of the filter inductor losses, an extensive loss model was constructed which had the actual currents and voltages from the simulation frameworks as inputs. The modeled core losses $P_{loss,core}$ are based on the improved generalized Steinmetz equation as presented in [23]. The wire losses $P_{loss,wire}$ are based on the methods presented in [24] to obtain the estimated skin effect losses (including dc losses) and proximity effect losses. The parameters used in the models correspond to the inductor as it was designed for the hardware prototype, which will be detailed in Section IV.

To estimate the losses generated in the flying capacitor $P_{loss,Cx}$ and filter capacitor $P_{loss,Cf}$, the currents through the respective capacitors were extracted from the simulation frameworks. The ESRs of the capacitors as implemented in the hardware prototype were then used to calculate the corresponding losses.

Each individual loss component was determined for a single period of the 100 Hz set-point current, of which the results are summarized in Table VII. Due to the high switching frequency on which both converter configurations operate, the switching losses $P_{loss,sw}$ and core losses $P_{loss,core}$ are relatively high. Together with the conduction losses in the switches $P_{loss,cond}$,

TABLE VII
OVERVIEW OF LOSSES FROM AC SIMULATIONS

Parameter	Two-level	Three-level	HS	Unit
$P_{loss,sw}$	38.9	12.4	44.3	W
$P_{loss,cond}$	36.2	18.8	9.85	W
$P_{loss,core}$	41.3	19.8	5.36	W
$P_{loss,wire}$	18.1	8.58	3.40	W
$P_{loss,Cx}$	0.149	0.430	0.270	W
$P_{loss,Cf}$	1.66	0.792	0.365	W
$P_{loss,total}$	136	60.8	63.5	W

these components dominate the loss breakdown. The almost two factor difference in losses between the two-level and three-level configurations is another important observation from the obtained results. Due to the lower rms and peak values of the current flowing through the switches, inductor, and filter capacitor, the corresponding losses are significantly smaller. Since the flying capacitor is actively delivering energy to the output in the three-level converter, the losses generated in the capacitor are higher. However, they remain insignificant compared to the losses generated in the switches and the inductor. From the loss breakdown analysis can be concluded that, applying three-level modulation results in a significant reduction in losses compared to two-level modulation.

The losses generated in the switches add up to 75.1 W for the two-level configuration and 31.2 W for the three-level configuration. In addition, simulation results were generated with three-level hard-switching modulation at $f_{sw} = 200$ kHz, of which the results are aggregated in Table VII in column HS. The rms and peak values of the filter inductor current for the hard-switching configuration are lower than that for both soft-switching configurations, resulting in lower conduction losses and lower core and wire losses in the inductor. The switching losses have however increased significantly due to the included turn-ON losses of the switches, resulting in total switch losses of 54.1 W. Subsequently, the total losses of the three-level soft-switching configuration come out lower than that for the three-level hard-switching modulation.

2) *Spectral Analysis*: High-precision industrial applications require a high linearity of the generated output current [2], [4], [5], [12]. In order to analyze the performance improvement with respect to linearity of the three-level converter configuration when compared to the two-level configuration, spectral analysis is performed by exciting the respective simulation frameworks with a 100-Hz sinusoidal set-point current. The amplitudes of the harmonics of the 100-Hz base frequency are calculated using the Fourier coefficients of a single period of the 100-Hz sine wave. The resulting values for the first 20 harmonics are shown in Fig. 14. The amplitudes of the third and fifth harmonic are higher for the two-level configuration when compared to the three-level configuration, whereas the amplitudes for almost all other harmonics are higher for the three-level configuration. However, the amplitude of the third harmonic for the two-level configuration is the dominant frequency component.

To quantify the nonlinear characteristics for both two- and three-level modulation, the spurious free dynamic range (SFDR)

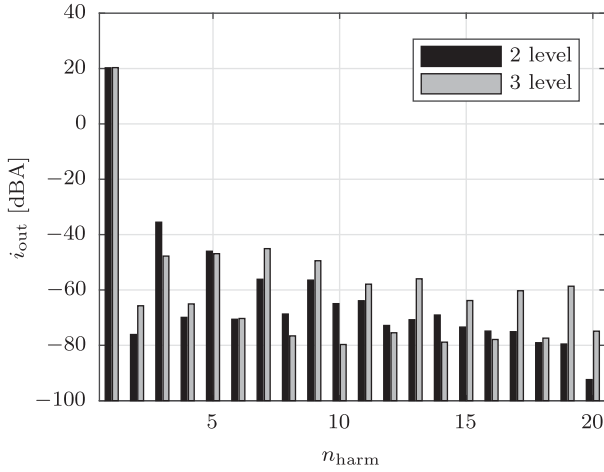


Fig. 14. Amplitudes of the first 20 harmonics of the 100-Hz base frequency calculated using Fourier coefficients of a single period of the 100-Hz sinusoidal set-point current for the two-level and three-level converter configurations.

TABLE VIII
CHARACTERISTICS FROM AC ANALYSIS

Parameter	Two-level	Three-level	Unit
SFDR	55.8	65.4	dB
THD	-55.4	-60.8	dB

and total harmonic distortion (THD) are calculated for the first 100 harmonics. The resulting values are gathered in Table VIII. From the spectral analysis can be concluded that, applying the three-level converter configuration results in an approximately 10-dB increase in SFDR and 5-dB decrease in THD. The performance of each converter configuration can be further improved by increasing the bandwidth of the output current control loop.

The position accuracy of a high-precision industrial application depends on the linearity of the output current generated by the power converter driving the actuators [4]. The increased linearity of the three-level converter configuration, as proposed by this article, with respect to the two-level converter configuration, which is based on an existing converter implementation, indicates potential performance improvement for a high-precision application.

IV. EXPERIMENTAL RESULTS

A hardware prototype setup was constructed of a three-level flying capacitor resonant pole inverter. The goal of the setup is to provide experimental results of the presented three-level modulation strategy, and validate the fast switching of high-voltage capabilities of the applied topology. A general description of the specifications of the prototype setup is given corresponding with the parameters used for the simulations performed in Section III. Measurements were performed to determine the actual required commutation charge Q_c , the delay incorporated in the zero-crossing detection, and dc and ac characteristics. The results of the measurements were used to improve the analytical description of the filter inductor current to better match the actual current.

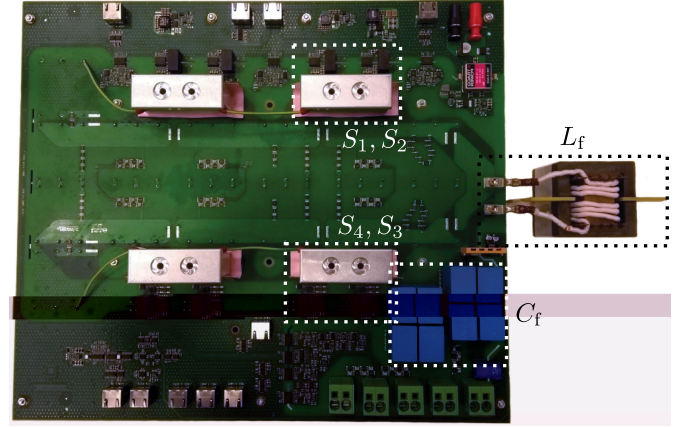


Fig. 15. Experimental setup with the four switches S_{1-4} , filter inductor L_f , and filter capacitor C_f indicated. The flying capacitor is located at the bottom of the PCB.

TABLE IX
SPECIFICATIONS OF PROTOTYPE SETUP

Parameter	Value	Unit
V_{dc}	1	kV
C_x	60	μ F
L_f	18.9	μ H
C_f	20	μ F
R_o	73.5	Ω
L_o	2.28	mH

A. Prototype Setup

The three-level prototype, of which the circuit schematic is shown in Fig. 2, was constructed using Wolfspeed C2M0025120D SiC MOSFETs as switching devices. The circuit parameters of the prototype setup, as shown in Fig. 15, are summarized in Table IX. The value for C_x was determined using the reasoning presented in Section II-C, and selected from the restricted set of available components that can withstand the flying capacitor voltage equal to V_{dc} plus a certain margin. The filter inductor consists of four E-cores (TDK E70/33/32 N87) with eight turns of Litze wire and 3-mm air gaps, resulting in a measured inductance of 18.9 μ H. A relatively large filter capacitor was chosen to limit the output voltage ripple. The resulting cutoff frequency of the L_f and C_f combination is around 8 kHz, leaving ample possibility to achieve the required bandwidth with sufficient phase margin.

The processor and field-programmable gate array (FPGA) of a dSpace Microlabbox were used to perform the calculation of the switching times and gate control, respectively. Closed-loop output current control was implemented for which the open-loop bandwidth was tuned to 1 kHz. The measured filter inductor current i_{L_f} was used for zero-crossing detection to synchronize the calculated switching times with the actual current. The control structure was set up according to the schematic overview given in Fig. 11. A differential voltage measurement circuit with a high CMMR was implemented to measure the flying capacitor voltage to facilitate active balancing of v_{C_x} .

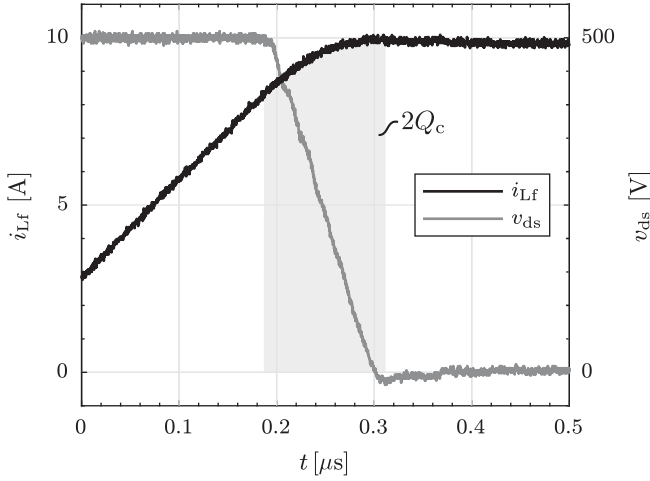


Fig. 16. Plot of the measurement results of the filter inductor current i_{Lf} (black) and the drain-source voltage v_{ds} (gray) of S_4 during commutation. The total commutation charge $2Q_c$ is indicated with a light gray area underneath i_{Lf} . The supply voltage for this measurement was set to $V_{dc} = 500$ V.

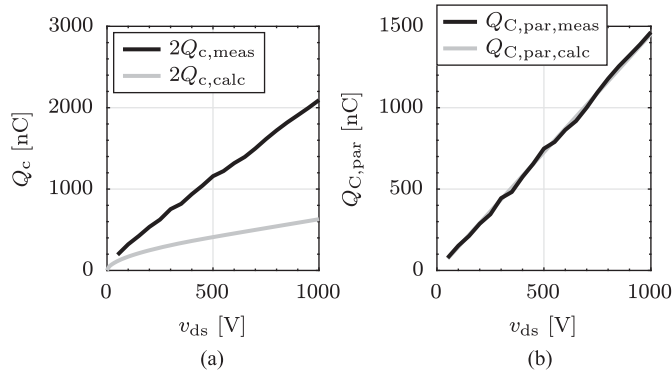


Fig. 17. (a) Plot of the measured (black) and calculated (gray) total commutation charges $2Q_c$ as a function of v_{ds} . (b) Plot of the measured (black) parasitic commutation charge $Q_{C,par,meas}$, obtained by subtracting the calculated total commutation charge $2Q_{c,calc}$ from the measured total commutation charge $2Q_{c,meas}$ for the same voltage. The calculated (gray) parasitic commutation charge $Q_{C,par,calc}$ represents a linear capacitance derived from $Q_{C,par,meas}$.

B. Determination of Actual Required Commutation Charge

As indicated in [18], additional parasitic capacitance of the switch node may have a significant influence on the total charge required for complete commutation of the switch node voltage. This parasitic capacitance is expected to manifest itself, for instance, between printed circuit board (PCB) traces, and the parasitic capacitance of the filter inductor. Measurements were performed to investigate the actual charge required for commutation. The total commutation charge $2Q_c$ was determined for a range of supply voltages V_{dc} by integrating the filter inductor current during commutation. An example of the resulting measurements of i_{Lf} and v_{ds} of S_4 , and the resulting total commutation charge $2Q_c$ for a supply voltage of $V_{dc} = 500$ V, is shown in Fig. 16. The measurement procedure was repeated for a supply voltage range of $V_{dc} = 50$ –1000 V, and the resulting total commutation charge $2Q_{c,meas}$ is shown in Fig. 17(a), together with the corresponding calculated values $2Q_{c,calc}$.

As can be seen in Fig. 17(a), a significant difference exists between the measured total commutation charge $2Q_{c,meas}$ and total commutation charge $2Q_{c,calc}$ calculated from the datasheet of the MOSFET (see Fig. 6). This indicates a significant parasitic capacitance present in the hardware prototype. When the calculated total commutation charge is subtracted from the measured total commutation charge for the same voltage, the charge in the additional parasitic capacitance of the switch node is obtained, designated $Q_{C,par,meas}$. Fig. 17(b) shows a plot of $Q_{C,par,meas}$, which approximately represents a straight line and therefore, indicates a linear capacitance. When the average capacitance of all measurements is calculated and used to determine the corresponding charge in that capacitance, the line designated $Q_{C,par,calc}$ in Fig. 17(b) is obtained. As can be concluded from the figure, the additional parasitic capacitance is indeed a linear capacitance, with value $C_{par} = 1.45$ nF. This result is in line with the findings presented in [18] regarding the linearity of additional parasitic capacitance of the switch node. Moreover, the significant increase in required commutation charge will influence the peak and rms values of the filter inductor current.

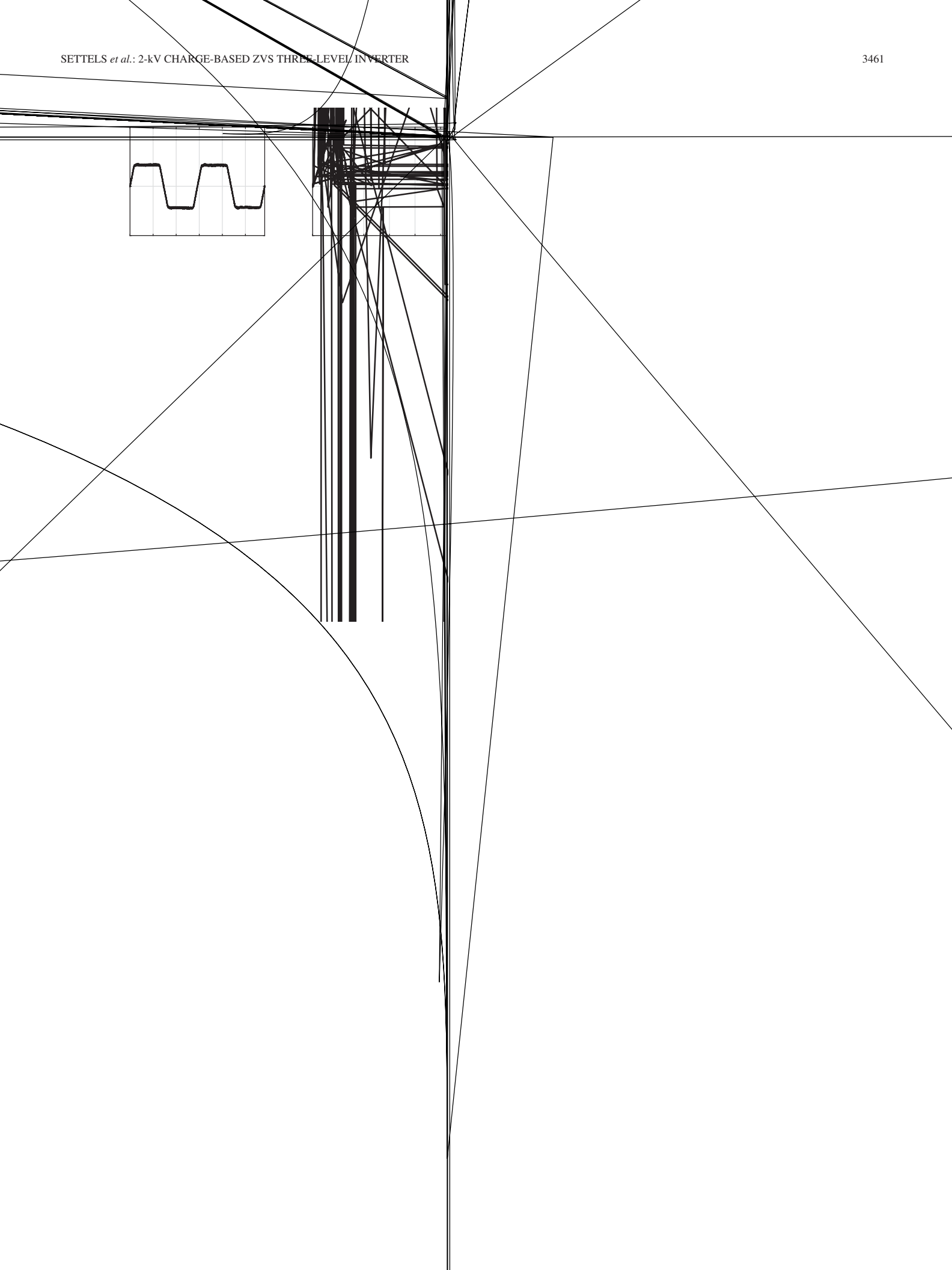
C. DC Measurements

To verify the proper functioning of the proposed charge-based ZVS with trapezoidal filter current, measurements were performed with a dc set-point current of $i_{set}^* = 0$ A and $i_{set}^* = 5$ A. The resulting waveforms of the filter inductor current i_{Lf} , switch-node voltage v_{sn} , flying capacitor voltage v_{Cx} , and output voltage v_{out} are shown in Fig. 18 for two switching periods.

The measured waveform of v_{sn} shows the distinct three-level modulation resulting in the trapezoidal shape of i_{Lf} . The waveforms of v_{Cx} for both set-point currents indicate proper balancing of the flying capacitor voltage around V_{dc} . However, voltage spikes of v_{Cx} are clearly visible at switching instants which increase in magnitude for higher set-point currents. The prototype contains relatively large parasitic inductance loops between S_2 , S_3 , and C_x , and between S_1 , C_x , S_4 , and V_{dc} . The parasitic inductances result in ringing of v_{sn} during commutation, of which the amplitude increases with increased dv_{sn}/dt , which in turn increases for higher i_{Lf} . The voltage spikes are visible in the waveforms of v_{out} and v_{sn} as well. The parasitic inductance can be greatly reduced by optimizing the PCB design to minimize the surface area of the loops.

To verify whether complete commutation of the switch-node voltage v_{sn} is achieved before turning ON a switch, the first commutation interval is investigated for $i_{set}^* = 0$ A. In Fig. 19, a plot of the waveforms of i_{Lf} , $v_{gs,2}$, $v_{gs,3}$, and $v_{ds,3}$ is shown for the interval corresponding with $[t_1, t_2]$ in Fig. 4. The plot shows that when $v_{gs,2}$ decreases to below the MOSFET threshold voltage, commutation of $v_{ds,3}$ from V_{dc} to 0 V is initiated. The waveform of $v_{gs,2}$ shows that switch S_3 is turned-ON when $v_{ds,3}$ is fully commutated and that zero-voltage turn-ON is achieved.

To verify complete commutation of the remaining intervals and for a nonzero set-point current, zoom-in plots are constructed around the commutation intervals as shown in Fig. 18. The filter inductor current i_{Lf} and switch-node voltage v_{sn} for the $i_{set}^* = 0$ A set point are shown in Fig. 20, and the waveforms for



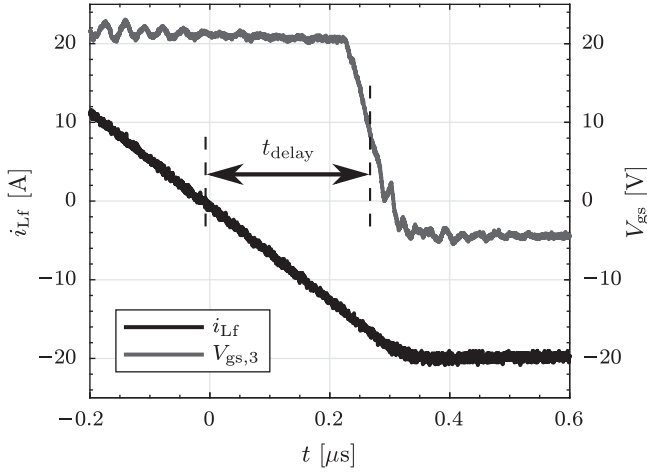


Fig. 22. Plots of the measured waveforms of i_{Lf} and $V_{gs,3}$ at a negative zero crossing of i_{Lf} . The time delay between the zero crossing of i_{Lf} and the turn-OFF instant of S_3 when $V_{gs,3}$ reaches the threshold voltage of the switch is indicated with t_{delay} .

end of the commutation interval increases significantly. This explains the voltage spikes shown in the plots of v_{Cx} and v_{out} of Fig. 18, which should remain limited to make sure the breakdown voltages of the components involved are not exceeded.

D. Compensation of Zero-Crossing Detection Delay

When comparing the measured waveforms of Fig. 18 with their equivalent obtained by simulation as shown in Fig. 12, the increased peak values for the filter inductor current stand out. The increased peak values originate to a small extent from the increased required commutation charge, as described in Section IV-B. However, the delay introduced in the detection of the zero crossing of the filter inductor current and the subsequent turn-OFF of the appropriate switch has a more significant impact. This is shown in Fig. 22, where the time delay t_{delay} between the zero crossing of i_{Lf} and the subsequent turn-OFF of S_3 is indicated. The total delay amounts to around $t_{delay} = 310$ ns which should be 0 s for this particular scenario where $i_{set}^* = 0$ A and $v_{out} = 0$ V. The main causes for the delay are the analog front-end of the Microlabbox and the gate driver. The impact can be mitigated in future designs by discarding the Microlabbox by using local control, and a gate driver with minimal propagation delay.

The increased required commutation charge Q_c in the hardware prototype can be easily implemented in the calculations of the switching times. The results shown in Fig. 18 indicate that piecewise linear approximation of i_{Lf} is still valid, especially for higher circulating currents. However, it is not possible to eliminate the significant delay introduced between the zero crossing of i_{Lf} and actual turn-OFF of a switch. Nonetheless, it is possible to take the delay into account when calculating the switching times by setting a minimum time for the intervals $[t_0, t_1]$ and $[t_5, t_6]$ as indicated in Fig. 4. The minimum time for both intervals is set equal to t_{delay} and the calculation of the switching times progresses accordingly.

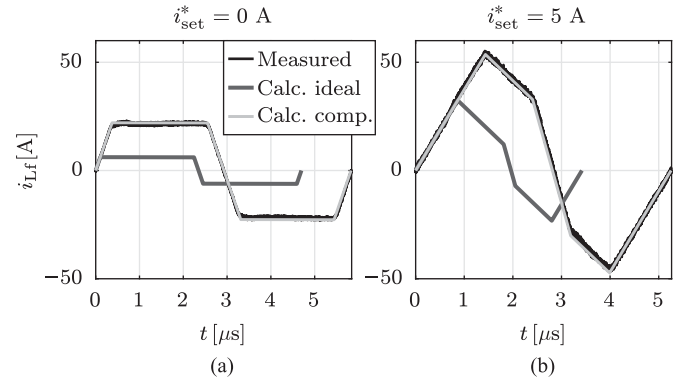


Fig. 23. Waveforms of i_{Lf} for a single switching period with (a) $i_{set}^* = 0$ A and (b) $i_{set}^* = 5$ A. The measured waveforms are plotted in black, the calculated ideal waveforms in dark gray, and the compensated calculated waveforms in light gray.

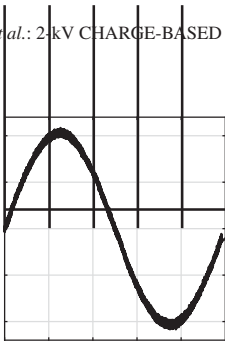
The resulting waveforms of i_{Lf} for the compensated calculations are shown in Fig. 23 for both $i_{set}^* = 0$ A and $i_{set}^* = 5$ A. The corresponding measured waveforms are shown in black, the ideal calculated waveforms in gray, and the compensated calculated waveforms in light gray. The significant differences between the ideal calculated waveforms and measured waveforms are illustrated again in the figure. Nonetheless, the compensated calculated waveforms, including the increased commutation charge $Q_{c,meas}$ and zero-crossing detection delay t_{delay} , show a significant correspondence with the measured waveform of i_{Lf} . Subsequently, the compensated calculated switching times are implemented in the hardware prototype and used to generate the measurement results.

E. AC Measurements

The ac performance of the three-level FC RPI was verified using a sinusoidal set point current with frequency $f_{i,set}^* = 100$ Hz and amplitude $|i_{set}^*| = 10$ A. The resulting waveforms of the output current i_{out} , output voltage v_{out} , filter inductor current i_{Lf} , and flying capacitor voltage v_{Cx} are shown in Fig. 24 for a single period of the 100-Hz sine wave.

The waveforms of i_{out} and v_{out} adopt the 100 Hz sinusoidal set point, with the addition of small spikes originating from the switching behavior, as was indicated in Fig. 18. The measured waveform of i_{Lf} shows similarities with its simulated counterpart shown in Fig. 13. However, due to the nonidealities explained in the previous paragraph, the envelope has a higher peak value. Moreover, a step in the envelope is clearly visible when the modulator switches between quadrants, due to the fact that the impact of the zero-crossing detection delay changes.

The waveform of v_{Cx} indicates proper balancing of the flying capacitor voltage around the reference voltage of 1 kV. The voltage spikes due to the switching behavior are visible in this plot as well, and are dominant with respect to the voltage ripple resulting from the voltage regulation. However, from the measured waveforms can be concluded that, a properly functioning converter is constructed.



simulation results. The divergence originated from an increased required commutation charge, and a delay between the zero crossing of the filter inductor current and the turn-OFF instant of a switch. Results obtained with compensated switching time calculations were presented, indicating a significant improvement. From the resulting measurements can be concluded that, fast switching of high voltage was achieved with a relatively high accuracy and linearity of the generated output current.

REFERENCES

- [1] H. Butler, "Position control in lithographic equipment [applications of control]," *IEEE Control Syst. Mag.*, vol. 31, no. 5, pp. 28–47, Oct. 2011.
- [2] J. M. Schellekens, "A class of robust switched-mode power amplifiers with high linear transfer characteristics," Ph.D. dissertation, Eindhoven Univ. Technol., Eindhoven, The Netherlands, 2014.
- [3] R. H. M. Schmidt, "Ultra-precision engineering in lithographic exposure equipment for the semiconductor industry," *Philos. Trans. A, Math. Phys. Eng. Sci.*, vol. 370, no. 1973, pp. 3950–3972, 2012. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pubmed/22802497>
- [4] S. J. Settels, J. van Duivenbode, and J. L. Duarte, "Impact of amplifier errors on position loop accuracy of high-precision moving stages," in *Proc. 19th Eur. Conf. Power Electron. Appl. (EPE'17 ECCE Eur.)*, Warsaw, Poland, Sep. 2017, pp. 1–10. [Online]. Available: <http://ieeexplore.ieee.org/document/8099009/>
- [5] M. Maurer, A. Tüysüz, and J. W. Kolar, "Distortion analysis of low-THD/high-bandwidth GaN/SiC class-D amplifier power stages," in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2015, pp. 2563–2571. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=7310020>
- [6] T. A. Meynard and H. Foch, "Multi-level conversion: High voltage choppers and voltage-source inverters," in *Proc. Rec. 23rd Annu. IEEE Power Electron. Spec. Conf.*, 1992, pp. 397–403. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=254717>
- [7] J. J. C. van Emden, J. van Duivenbode, and J. L. Duarte, "Flying capacitor resonant pole inverter topology with reduced switch voltage stress," *Eur. Power Electron. Drives J.*, vol. 24, no. 4, pp. 6–13, 2014.
- [8] S. J. Settels, J. Everts, and J. van Duivenbode, "Charge-based zero-voltage switching of a flying capacitor resonant pole inverter with trapezoidal filter current," in *Proc. 42nd Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2016, pp. 3282–3287. [Online]. Available: <http://ieeexplore.ieee.org/document/7793283/>
- [9] J. M. Schellekens, J. L. Duarte, M. A. M. Hendrix, and H. Huisman, "Interleaved switching of parallel ZVS hysteresis current controlled inverters," in *Proc. Int. Power Electron. Conf. – ECCE Asia*, Jun. 2010, pp. 2822–2829. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=5543891>
- [10] D. M. Divan and G. Skibinski, "Zero-switching-loss inverters for high-power applications," *IEEE Trans. Ind. Appl.*, vol. 25, no. 4, pp. 634–643, Jul./Aug. 1989.
- [11] J. Cho, D. Hu, and G. Cho, "Three phase sine wave voltage source inverter using the soft switched resonant poles," in *Proc. 15th Annu. Conf. IEEE Ind. Electron. Soc.*, 1989, pp. 48–53. [Online]. Available: <http://ieeexplore.ieee.org/document/69610/>
- [12] S. J. Settels, J. L. Duarte, J. van Duivenbode, and C. G. E. Wijnands, "High voltage power amplifier for high precision mechatronic systems," in *Proc. 20th Eur. Conf. Power Electron. Appl.*, Riga, Latvia, Sep. 2018, pp. 1–10.
- [13] J. van Duivenbode and B. Smet, "An empiric approach to establishing mosfet failure rate induced by single-event burnout," in *Proc. 13th Int. Power Electron. Motion Control Conf.*, Sep. 2008, pp. 102–107. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=4635251>
- [14] R. W. de Doncker and J. P. Lyons, "The auxiliary resonant commutated pole converter," in *Proc. Conf. Rec. IEEE Ind. Appl. Soc. Annu. Meeting*, 1990, pp. 1228–1235. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=152341>
- [15] C. Marxgut, F. Krismer, D. Bortis, and J. W. Kolar, "Ultraflat interleaved triangular current mode (TCM) single-phase PFC rectifier," *IEEE Trans. Power Electron.*, vol. 29, no. 2, pp. 873–882, Feb. 2014. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=6504772>
- [16] H. Obara and Y. Sato, "Theoretical analysis of self-balancing function of capacitor voltages in flying capacitor multi-level converters," in *IEEE Energy Convers. Congr. Expo.*, Sep. 2011, pp. 2079–2086. [Online]. Available: <http://ieeexplore.ieee.org/document/6064043/>
- [17] C. Marxgut, J. Biela, and J. W. Kolar, "Interleaved triangular current mode (TCM) resonant transition, single phase PFC rectifier with high efficiency and high power density," in *Proc. Int. Power Electron. Conf. – ECCE Asia*, Jun. 2010, pp. 1725–1732.
- [18] M. Kasper, R. M. Burkart, G. Deboy, and J. W. Kolar, "ZVS of power mosfets revisited," *IEEE Trans. Power Electron.*, vol. 31, no. 12, pp. 8063–8067, Dec. 2016. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=7482851>
- [19] "Datasheet: C2M0025120D silicon carbide power mosfet," Wolfspeed, Cree, Inc., Durham, NC, USA. [Online]. Available: https://www.wolfspeed.com/downloads/dl/file/id/161/product/8/c2m0025120_d.pdf
- [20] S. J. Settels, J. van Duivenbode, J. L. Duarte, and E. A. Lomonova, "Flying capacitor resonant pole inverter applying five voltage levels," in *Proc. IEEE Energy Convers. Congr. Expo.*, Oct. 2017, pp. 2121–2128. [Online]. Available: <http://ieeexplore.ieee.org/document/8096420/>
- [21] S. Tiwari, J. K. Langelid, O.-M. Midtgard, and T. M. Undeland, "Soft switching loss measurements of a 1.2 kV SiC MOSFET module by both electrical and calorimetric methods for high frequency applications," in *Proc. 19th Eur. Conf. Power Electron. Appl.*, Sep. 2017, pp. 1–10. [Online]. Available: <http://ieeexplore.ieee.org/document/8099207/>
- [22] S. Tiwari, J. K. Langelid, O.-M. Midtgard, and T. M. Undeland, "Hard and soft switching losses of a SiC MOSFET module under realistic topology and loading conditions," in *Proc. 19th Eur. Conf. Power Electron. Appl.*, Sep. 2017, pp. 1–10. [Online]. Available: <http://ieeexplore.ieee.org/document/8099140/>
- [23] J. Mühlethaler, J. Biela, J. W. Kolar, and A. Ecklebe, "Improved core-loss calculation for magnetic components employed in power electronic systems," *IEEE Trans. Power Electron.*, vol. 27, no. 2, pp. 964–973, Feb. 2012. [Online]. Available: <http://ieeexplore.ieee.org/document/5955126/>
- [24] J. Mühlethaler, J. W. Kolar, and A. Ecklebe, "Loss modelling of inductive components employed in power electronic systems," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2011, pp. 945–952.



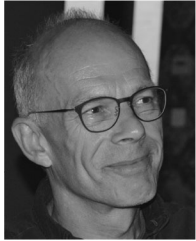
Sjeff J. Settels (S'15) received the M.Sc. degree in electrical engineering from the Eindhoven University of Technology, Eindhoven, The Netherlands, in 2012, with a focus on power electronics and signal processing. He is currently working toward the Ph.D. degree at the Electromechanics and Power Electronics Group of the same university.

Following his graduation, he was with ASML, a semiconductor lithography company. His current research interests include multilevel topologies for high-power high-accuracy power electronics.



Jorge L. Duarte (M'99) received the M.Sc. degree from the University of Rio de Janeiro, Brazil, and the Dr.-Ing. degree from the INP Lorraine, Nancy, France, in 1980 and 1985, respectively.

Since 1990, he has been an Assistant Professor (UD) with the Electromechanics and Power Electronics Group of the Department of Electrical Engineering, Eindhoven University of Technology, Eindhoven, The Netherlands. In 1989, he was ap-



Jeroen van Duivenbode (M'13) received the M.Sc. degree in power electronics and avionics from Delft University, Delft, the Netherlands, in 1987.

After the completion of his studies, he has developed satellite electronics during employments in Toulouse, France, and Horten, Norway. In 1998, he joined the semiconductor lithography company ASML, Veldhoven, The Netherlands. Since 2012, he has been a Part-Time Research Fellow with the Department of Electrical Engineering, Eindhoven University of Technology, Eindhoven, The Netherlands, and was appointed as a "TU/e Fellow" in 2014. He holds a patent on vacuum high-voltage connectors and has published on the subject of power device failures in terrestrial applications due to cosmic rays. His research interests include the development of highly accurate and reliable power electronics.



Elena A. Lomonova (M'04–SM'07) graduated (cum laude) in electromechanical and control systems from Moscow Aviation Institute (State University of Aerospace Technology), Moscow, Russia, and received the Ph.D. degree (cum laude) in 1993, working on powertrain and control systems for autonomous vehicles with multilevel power supply subsystems for on-board loads and laser equipment.

She started her industrial career with the research and development company "Astrophysics," Moscow, Russia (1982–1987). Afterward, she moved to the Electromechanical and Control Systems Department, State University of Aerospace Technology (MAI), and was active in research, education, and industrial projects (1987–1997). In 1998, she was with the Delft University of Technology and joined Eindhoven University of Technology in 2000. In March 2009, she was appointed a Full-Time Professor. She is currently a Full Professor and Chair of the Electromechanics and Power Electronics Group. Her chair focuses on fundamental and applied research on enabling energy conversion theory, methods and technologies for high-precision, automotive, and medical systems. She is an author and co-author of more than 450 scientific publications and more than ten patents. Her research interests include various facets of advanced mechatronics, electromechanics, and electromagnetics, including rotary electrical machines and drives, and linear and planar actuation systems.